

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2021

ESE 3107

(Solar Thermal Engineering)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | What is sunshine hour? How it is related to declination? | 07 |
| 1(b). | Define the following terms with schematic diagram:
(i) Zenith angle, (ii) Solar altitude angle, and (iii) Solar azimuth angle. | 09 |
| 1(c). | Distinguish between extraterrestrial and terrestrial radiation. | 06 |
| 1(d). | Determine the sunshine hour for Khulna city on January 18 (latitude and longitude of Khulna city are 22.80 N and 89.24 E) | 07 |
| 1(e). | What is meant by clearness index and diffuse fraction? | 06 |
| 2(a). | Explain the working principle of pyranometer. How beam radiation can be measured using it? | 12 |
| 2(b). | Explain the incident solar radiation on tilted surface. How beam radiation in tilted surface can be calculated using geometric factor? | 08 |
| 2(c). | What is meant by black body? State and explain Stefan-Boltzmann law. | 09 |
| 2(d). | State the Kirchhoff's law. | 06 |
| 3(a). | What is view factor? State the reciprocity relationship. | 07 |
| 3(b). | Derive the expression for net radiation exchange between two infinitely long parallel plates using network method. | 10 |
| 3(c). | What is meant by radiation shield. Why it is used? | 05 |
| 3(d). | Two large, opaque, gray, diffusely emitting and reflecting parallel plates have emissivities $\epsilon_1 = 0.7$ and $\epsilon_2 = 0.5$ and are maintained at uniform temperatures $T_1 = 900$ K and $T_2 = 500$ K. Calculate the net radiation exchange between them. | 13 |
| 4(a). | Explain the construction details of flat plate solar thermal collector with schematic diagram. | 13 |
| 4(b). | What are the advantages and limitations of evacuated tube solar collectors. | 10 |
| 4(c). | Explain the working principle of an evacuated tube solar collector with a heat pipe. | 12 |

SECTION – B

- 5(a). Draw the thermal network for a double glass covered FPC. 10
- 5(b). Compare the materials used for glazing and absorber. 06
- 5(c). Determine the overall heat transfer coefficient for a FPC system inclined at 45° to the horizontal and facing due south for the following parameters: 19
- The average ambient air temperature for the day is 20°C
 - The observed glass and absorber plate temperature are 50°C and 70°C , respectively
 - The system is provided with 5 cm thick insulation (glass wool) at the bottom and the thermal conductivity of insulation is $0.03 \text{ W/m}^\circ\text{C}$
 - The air space between the absorber plate and the glass cover has an optimum thickness of 75 cm
 - The emissivity of the glass and the plate are 0.8 and 0.95, respectively.
- 6(a). Define the followings: 10
(i) Acceptance angle, (ii) Concentration ratio, (iii) Critical radiation level, (iv) Collector efficiency factor, (v) Collector heat removal factor
- 6(b). With necessary figure, classify the solar concentrator. 14
- 6(c). Why tracking is necessary for a concentrating collector? What are the advantages and disadvantages of concentrating collector? 11
- 7(a). Explain the working principle of an indirect solar water heating system. 10
- 7(b). Explain the integrated collector storage system. What are its disadvantages? 08
- 7(c). How to overcome the limitations of open sun crop drying by using solar thermal systems? 05
- 7(d). Explain the working principle of solar assisted desiccant cooling system. 12
- 8(a). Write down the differences between solar pool heating and solar pond system. 08
- 8(b). How solar collectors can be used to generate electricity? With the help of schematic diagram, explain the parabolic dish system to generate energy at high temperature. 14
- 8(c). What is solar desalination? 13
- A solar still have water and glass temperature equal to 50°C and 40°C , respectively. If the convective heat transfer coefficient from water surface to glass is $2.40 \text{ W/m}^2\text{K}$, estimate the hourly distillate output per square meter from the solar still.

--- End ---

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2021

EE 3113

(Power System Engineering)

Time: 3 Hours.

Full Marks: 210

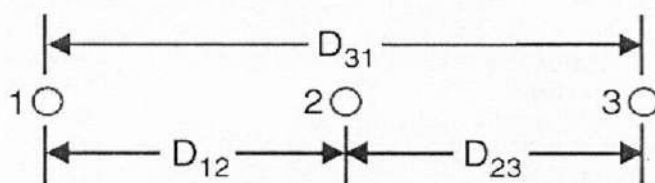
N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

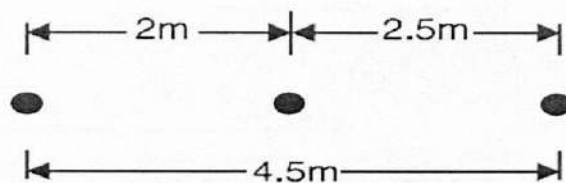
SECTION – A

- 1(a). What is Skin effect and Transmission efficiency? Why skin effect is absent in the d.c. system? 06
- 1(b). What is meant by the parametres of over head transmission lines? Derive an expression for the inductance of composite-conductor lines. 14
- 1(c). What is the purpose of an overhead transmission line? 05
- 1(d). Calculate the inductance of each conductor in a 3-phase, 3-wire system when the conductors are arranged in a horizontal plane with spacing such that $D_{31} = 4\text{ m}$; $D_{12} = D_{23} = 2\text{ m}$ as shown in figure. The conductors are transposed and have a diameter of 2.5 cm. 10



- 2(a). Why inductance and capacitances are formed in transmission line? Show that capacitance of a 3-phase line with equilateral spacing is $C_n = \frac{2\pi K}{\ln(D/r)}$; where the symbols have their usual meaning. 15
- 2(b). What is electric power supply system? Draw a single line diagram of a typical a.c. power supply scheme. 06
- 2(c). Illustrate the Nominal T Method and π Method for transmission line. 06
- 2(d). A 3-Phase, 50Hz, 66 kV overhead line conductors are placed in a horizontal plane as shown in following figure. The conductor diameter is 1.25 cm. If the line length is 100 km and assuming complete transposition of the line, calculate the followings: 08

(i) capacitance per phase, (ii) charging current per phase,

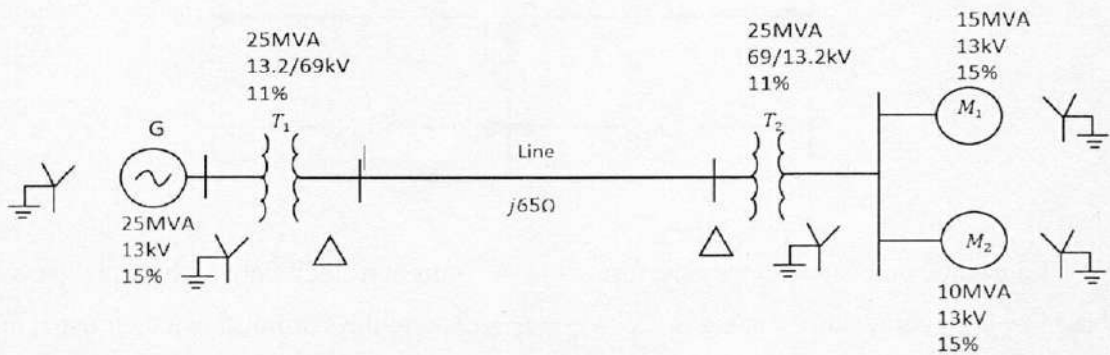


- 3(a). Explain the following terms with reference to corona: 06
- i) Critical disruptive voltage
 - ii) Visual critical voltage
 - iii) Power loss due to corona
- 3(b). Given reasons for unequal potential distribution over a string of suspension insulators? Show that in a string of suspension insulators, the disc nearest to the conductor has the highest voltage across it. 14
- 3(c). Why are insulators used with overhead lines? Discuss the desirable properties of Insulators. 07
- 3(d). Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is $\frac{1}{8}$ th of the capacitance of the insulator itself. Also find the string efficiency. 08

- 4(a). What is meant by sag? Deduce an approximate expression for sag in overhead lines when supports are unequal level. 12
- 4(b). A transmission line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2 . The tension in the conductor is 2000 kg. If the specific gravity of the conductor material is 9.9 gm/cm^3 and wind pressure is 1.5 kg/m length, calculate the sag. Also calculate the vertical sag. 12
- 4(c). Derive an expression for the flux linkages and inductance between two points external to an isolated conductor. 11

SECTION – B

- 5(a). What is P.U. system? Write the benefits of P.U. system for analyzing the power system. 10
- 5(b). Show that the equivalent impedance of a transformer in per unit referred to the LV and HV side is same. 10
- 5(c). A one-line diagram of a three-phase power system is shown in figure. Choose 13.8 kV as the base voltage for the generator voltage, and 25 MVA as the base MVA. Draw a per unit reactance diagram. 15



- 6(a). Describe the importance of load flow analysis in power systems. Classify various types of buses for load flow studies. 10
- 6(b). What are the various methods of voltage control in power system? Describe the on-load tap changing transformer method of voltage control. 10
- 6(c). A three-bus power system is shown in the following figure. The relevant per unit line admittances on 100MVA base are indicated on the diagram and bus data are given in the table. From Y_{bus} , determine the voltage at bus 2 and 3 after the second iteration using Gauss-Seidel method. Take the acceleration factor $\alpha = 1.6$. 15

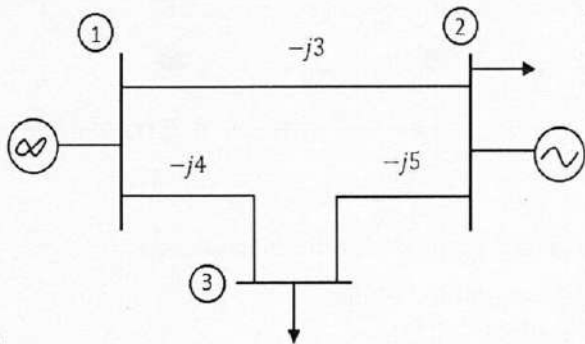


Table: Bus data

Bus number	Type	Generation		Load		Bus voltage	
		P_G (MW)	Q_G (MVARr)	P_L (MW)	Q_L (MVARr)	V (PU)	δ (deg)
1	slack	?	?	0	0	1.02	0°
2	PQ	25	15	50	25	?	?
3	PQ	0	0	60	30	?	?

7(a).	What is a fuse? Mention the describe characteristics of fuse element.	06
7(b).	Define and explain the followings terms as applied to circuit breakers: (i) Arc voltage, (ii) Restriking voltage, (iii) Recovery voltage.	09
7(c).	Explain the construction and principle of operation of Sulphur Hexafluoride (SF_6) circuit breaker.	10
7(d).	What is lightning? Describe how a lightning arrester conducts the high voltage surges on the power system to the ground.	10
8(a).	Describe the causes of faults in power systems.	05
8(b).	Define symmetrical components. Show that the total complex power in an unbalanced system is equal to the sum of the complex powers of the three symmetrical components.	10
8(c).	Derive an expression for fault current for single line-to-ground faults by using the concept of symmetrical components and draw the interconnection of sequence networks for the single line-to-ground fault.	12
8(d).	What is smart grid? Discuss the key functions of smart grid with smart grid's communication system.	08

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2021

ESE 3123

(Thermo-fluid Devices)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- | | | |
|-------|---|----|
| 1(a). | What are the different kinds of losses in pipe flow? Briefly discuss about it. | 10 |
| 1(b). | State and discuss Bernoulli's equation. How it can be derived from Euler's equation? | 10 |
| 1(c). | Two pipes each 300 m long are available for connecting to a reservoir from which a flow of $0.085 \text{ m}^3/\text{s}$ is required. If the diameters of the two pipes are 0.30 m and 0.15 m, respectively, determine the ratio of the head lost when the pipes are connected in series and connected in parallel. Neglect minor losses. | 15 |
| 2(a). | State and discuss Buckingham- π theorem. How it differs from Rayleigh's method? | 08 |
| 2(b). | What are hydraulic coefficients? Deduce an expression of measuring actual velocity at any point of a flowing fluid by using a Pitot tube. | 10 |
| 2(c). | Briefly discuss about vena-contracta. | 04 |
| 2(d). | The pressure difference ΔP in a pipe of diameter D and length l due to turbulent flow depends on the velocity V , viscosity ν , density ρ and roughness k . Using Buckingham's- π theorem, obtain an expression for pressure difference ΔP . | 13 |
| 3(a). | What is meant by hydraulic machines? Write down the basic difference between a turbine and a pump. | 05 |
| 3(b). | Define centrifugal pump. What are the main parts of a centrifugal pump. Briefly discuss with the help of necessary sketches. | 10 |
| 3(c). | Define the following terms:
(i) Mechanical efficiency, (ii) Manometric head, (iii) static head, and (iv) manometric efficiency. | 08 |
| 3(d). | A single-stage centrifugal pump with impeller diameter of 50 cm rotates at 1800 rpm and lifts 4 m^3 of water per second to a height of 25 m with an efficiency of 70%. Find the number of stages and diameter of each impeller of a similar multistage pump to lift 5 m^3 of water per second to a height of 180 m when rotating at 1300 rpm. | 12 |
| 4(a). | What is priming? Why is it necessary? | 07 |
| 4(b). | Derive an expression for measuring the maximum suction height of a centrifugal pump. | 10 |
| 4(c). | Write down the classifications of reciprocating pump. Also differentiate between centrifugal pump and reciprocating pump. | 08 |
| 4(d). | A double-acting reciprocating pump running at 40 rpm, is discharging 2 m^3 of water per minute. The pump has a stroke of 500 mm. The diameter of the piston is 250 mm. The delivery and suction head are 18 m and 5 m, respectively. Find the slip of the pump and power required to drive the pump. | 10 |

SECTION – B

- 5(a). What is meant by compressible flow? Differentiate between compressible and incompressible flow. 06
- 5(b). Show mathematically that a converging-diverging duct can be used as a nozzle or diffuser. 11
- 5(c). What are meant by critical properties and critical ratios? Briefly discuss the significance of them. 04
- 5(d). Air is flowing through a duct when the stagnation pressure and temperature are 100 kPa absolute and 35°C, respectively. If the static pressure of air is 52 kPa absolute, calculate the velocity of air considering compressibility effects and ignoring compressibility effects. Take $k = 1.4$ and $R = 287 \text{ J/kg.K}$. 14
- 6(a). Prove that: $\dot{M}_{\max} = A \times P_0 \sqrt{\frac{k}{RT_0} \left(\frac{2}{k+1} \right)^{(k+1)/[2(k-1)]}}$ 16
- Where the symbols have their usual meanings.
- 6(b). Differentiate between normal shock wave and oblique shock wave. 05
- 6(c). Air enters a converging-diverging nozzle at 1 MPa and 300 K with a low velocity. If a normal shock wave occurs at the exit plane of the nozzle at $M_a = 2$, determine the pressure, temperature, Mach number and velocity after the shock wave. 14
- 7(a). Show that the heat transfer rate in a parallel-flow heat exchanger can be given by: 15
- $\dot{Q} = UA_s \Delta T_{lm}$
- Where the symbols have their usual meanings.
- 7(b). What is the physical significance of NTU method? 05
- 7(c). Engine oil is to be cooled from 80°C to 50°C by using a single-pass, counter flow, concentric-tube heat exchanger with cooling water available at 20°C. Water flows inside a tube with an ID of $D_i = 2.5 \text{ cm}$ at a rate of $\dot{m}_w = 0.08 \text{ kg/s}$, and oil flows through the annulus at a rate of $\dot{m}_{oil} = 0.16 \text{ kg/s}$. The heat transfer coefficients for the water side and oil side are $h_w = 100 \text{ W/m}^2.\text{°C}$ and $h_{oil} = 80 \text{ W/m}^2.\text{°C}$, respectively. The fouling factors are $F_w = 0.00018 \text{ m}^2.\text{°C/W}$ and $F_{oil} = 0.00018 \text{ m}^2.\text{°C/W}$; and the tube wall resistance is negligible. The specific heats of water and oil are $c_w = 4180 \text{ J/kg.°C}$ and $c_{oil} = 2020 \text{ J/kg.°C}$, respectively. Calculate the tube length required. 15
- 8(a). Define heat exchanger. Write down the classification of heat exchanger. 06
- 8(b). What is meant by heat effectiveness of a heat exchanger? Show that the overall heat transfer coefficient for a double-pipe heat exchanger is given by: 14
- $$U = \frac{1}{\frac{1}{h_i} + \frac{1}{h_o}}$$
- Where the symbols have their usual meanings.
- 8(c). A counter flow heat exchanger is employed to cool 0.55 kg/s ($c_p = 2.45 \text{ kJ/kg.°C}$) of oil from 115°C to 40°C by the use of water. The inlet and outlet temperatures of cooling water are 15°C and 75°C, respectively. The overall heat transfer coefficient is expected to be 1450 W/m².°C. Using NTU method, calculate the followings: 15
- (i) The mass flow rate of water;
 - (ii) The effectiveness of the heat exchanger; and
 - (iii) The surface area required.

--- End ---

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2021

ESE 3105

(Heat and Mass Transfer)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

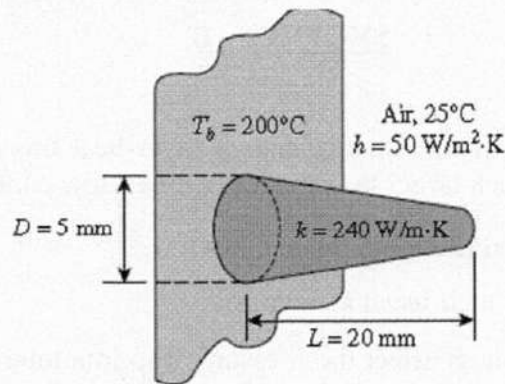
ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

iv) Table, chart may be provided on request.

SECTION – A

- 1(a). Explain various kinds of boundary conditions commonly considered in heat conduction problems. 10
- 1(b). Does heat generation in solid violates the first law of thermodynamics? Explain. 08
- 1(c). Consider a long resistance wire of radius $r_1 = 0.2$ cm and thermal conductivity $k_{\text{wire}} = 15$ W/m.K in which heat is generated uniformly as a result of resistance heating at a constant rate of 50 W/cm³. The wire is embedded in a 0.5 cm thick layer of ceramic whose thermal conductivity is $k_{\text{ceramic}} = 1.2$ W/m.K. If the outer surface temperature of the ceramic is measured to be $T_s = 45^\circ\text{C}$, find the centerline temperature and the interface temperature under steady condition. 17
- 2(a). Define heat transfer augmentation. Why is it necessary? 05
- 2(b). Explain the concept of critical thickness of insulation. Derive an expression for the critical radius of insulation when it is put outside of a small hollow pipe. 15
- 2(c). Aluminium pin fins of parabolic profile with blunt tips are attached on a plane wall with surface temperature of 200°C . Each fin has a length of 20 mm and a base diameter of 5 mm. The fins are exposed to an ambient air condition of 25°C and the convective heat transfer coefficient is 50 W/m².K. If the thermal conductivity of the fins is 240 W/m.K, determine the efficiency and heat transfer rate of each fin. 15



- 3(a). A semi-infinite solid having an uniform initial temperature T_i (at time $t = 0$) and then subjected to a constant temperature T_s at the boundary $x = 0$. Obtain the following nondimensional temperature distribution equation when $t > 0$, by using similarity variable (η). 17

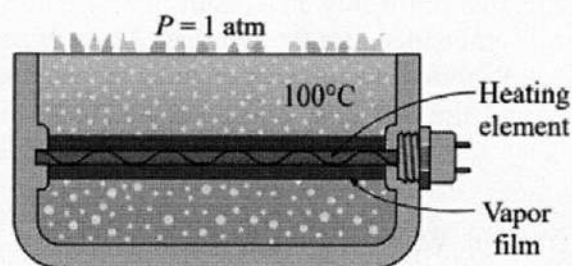
$$\frac{T - T_s}{T_i - T_s} = 1 - \text{erfc}(\eta)$$

- 3(b). Two semi-infinite solids A and B of constant thermophysical properties initially being at uniform temperature $T_{A,i}$ and $T_{B,i}$ are brought into perfect contact. Show that the steady state interface temperature of these two bodies in contact T_s can be given as: 10

$$T_s = \frac{T_{A,i} + \alpha T_{B,i}}{1 + \alpha}; \text{ where } \alpha = \frac{\sqrt{(k\rho c_p)_B}}{\sqrt{(k\rho c_p)_A}}$$

Where the symbols have their usual meanings.

- 3(c). The ground at a particular location is covered with snowpack at -10°C for a continuous period of 60 days and the average soil properties at that location are $k = 0.4 \text{ W/m.K}$ and $\alpha = 0.15 \times 10^{-6} \text{ m}^2/\text{s}$. Assuming an initial uniform temperature of 15°C for the ground, determine the minimum burial depth to prevent the water pipes from freezing. 08
- 4(a). Distinguish between film-wise and drop-wise condensation. Which one is more effective mechanism of heat transfer? Discuss. 08
- 4(b). Draw the boiling curve and identify different boiling regimes. Also explain the characteristics of each regime. 15
- 4(c). Water is boiled at atmospheric pressure by a horizontal polished copper heating element of diameter 5 mm and emissivity of 0.05 immersed in water as shown in figure. If the surface temperature of the heating wire is 350°C , determine the rate of heat transfer from the wire to the water per unit length of the wire. 12



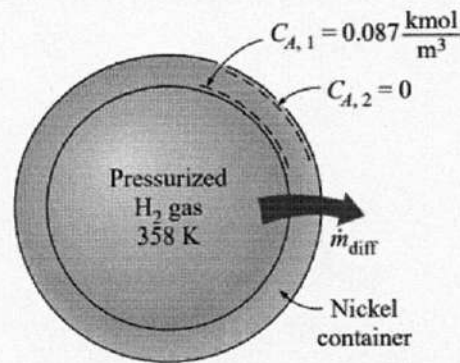
[Following empirical correlations can be used]

$$q_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 C_{pv} (T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}})$$

SECTION – B

- 5(a). Explain the phenomena of turbulent boundary layer heat transfer. Why turbulent heat transfer coefficient is much larger than that of laminar flow condition? 13
- 5(b). For forced convection inside a tube, prove that $st = \frac{f}{8}$. 12
Where the symbols have their usual meanings.
- 5(c). How does surface roughness affect the pressure drop in a tube if the flow is turbulent? What would be your opinion if the flow were laminar? 05
- 5(d). With a neat sketch show the hydrodynamic and thermal entry lengths for a fluid flowing through a tube having Prandtl number of 0.7 and 7, respectively. 05
- 6(a). Using dimensional analysis, show that $Nu = f(Re, Pr)$ for forced convection heat transfer over a flat plate. 16
- 6(b). Write down the significance of Grashof number. 04
- 6(c). In a long annulus tube (35 mm ID and 50 mm OD), water is heated by maintaining the outer surface of inner tube at 60°C . Water enters at 20°C and leaves at 34°C , while its flow velocity is 2 m/s. Estimate the heat transfer coefficient. 15
Following empirical relations can be used:
 $Nu = 0.023 Re^{0.8} Pr^n$ for $Re > 10000$ and $0.7 \leq Pr \leq 160$
 $n = 0.4$ for heating and $n = 0.3$ for cooling
- 7(a). Prove that the axial temperature gradient does not depend on radial flow direction in case of uniform surface heat flux condition for a thermally fully developed flow. 15

- 7(b). Air at atmospheric pressure and temperature of 325 K flows through a tube bundle in in-line tube arrangement. Tubes have an outside diameter of 1.9 cm and are maintained at a uniform temperature of 375 K. The longitudinal and transverse pitches for the bundle are given by $\frac{S_L}{D} = \frac{S_T}{D} = 2$. The bundle consists of 15 rows, 20 tubes per row and 0.75 m long in the direction of flow. The air velocity just before the tube bank is 8 m/s. 20
- Find the pressure drop across the tube bundle.
 - Find the exit temperature of the air.
 - Determine the total heat transfer rate.
- 8(a). Write down the differences between absorption and adsorption. Draw the schematic diagram of a simple distillation process. 10
- 8(b). State and describe Fick's law of diffusion. How does the mass diffusivity of a gas mixture change with (i) temperature, and (ii) pressure? 10
- 8(c). Pressurized hydrogen gas is stored at 358 K in a 4.8 m outer diameter spherical container made of nickel shown in figure. The shell of the container is 6 cm thick. The molar concentration of hydrogen in the nickel at the inner surface is determined to be 0.087 kmol/m^3 . The concentration of hydrogen in the nickel at the outer surface is negligible. Determine the mass flow rate of hydrogen by diffusion through the nickel container. 15



--- END ---

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY**Department of Energy Science and Engineering**

B. Sc. Engineering 3rd Year 1st Term Examination, 2021

Math 3113

(Statistics and Numerical Methods)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Write down Bayes' theorem. A hotel gets cars for its guest from three rental agencies, 20% from agency X, 35% from agency Y and 45% from agency Z. If 10% of the car from X, 6% of the car from Y and 12% of the car from Z need tune-ups, what is the probability that a car need tune-up which is delivered to a guest of the hotel came from rental agency Z? 09
- 1(b). Discuss the different types of frequency curves. 10
- 1(c). An industry has two factories A and B. The average weekly wages and standard deviation of the two factories as follows: 16

Factory	Average wages	Standard deviation	No. of worker
A	450	45	110
B	600	50	80

- (i) In which factory is there greater variation in the distribution of weekly wages?
(ii) Find the combined mean and standard deviation.

- 2(a). Find first four moments measured from origin. Also find first four central moments. Using above moments find means, standard deviation and moment coefficient of skewness. Also comment about the distribution; where 20

error	6-10	11-15	16-20
counts	5	8	5

- 2(b). What is meant by the dispersion? From the following frequency distribution, compute the mean and standard deviation of 100 students: 10

Mass (in kg)	60-62	63-65	66-68	69-71	72-74
Number of students	5	18	42	27	8

- 2(c). State the postulates of the probability. 05

- 3(a). There are two factories X and Y; the average daily wages and standard deviation of the two factories are given below: 12

Factory	Average wages	Standard deviation	No. of workers
X	45	5	11
Y	60	10	8

Suppose in factory B, the daily wages of a worker was wrongly posted as Tk 70 instead of Tk 40. What is the current mean and variance of factory B?

- 3(b). Test whether the following functions are probability distribution or not. 13

(i)

x	-1	0	1	2
P(x)	1	3k	2k	0.4

- (ii) $f(x) = \begin{cases} 1/6 & \text{when } 1 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$

Hence, if possible, find $E(2x^2 + 3)$ standard deviation, moment coefficient of skewness and $P(-1 \leq x \leq 2)$ only for problem (ii).

- 3(c). Define the root mean square (RMS). Determine the RMS of the reaction times of an individual to contain stimuli were measured by a psychologist to be 0.53, 0.46, 0.50, 0.49, 0.52, 0.53, 0.44, and 0.55 second respectively. 10

- 4(a). Fit the following data to Binomial Distribution. Hence test the goodness of fit with 5% level of significance. Three coin are tossed and output are shown in the following table 14

No. of Head	0	1	2	3
Observations	2	3	2	1

- 4(b). The number of arrivals of customers during any day follows the Poisson distribution with a mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2? 10

- 4(c). In a new school, the mean and variance of 500 students' IQ score are 40 and 16 respectively. Find the expected number of students whose score are 11

- (i) lies between 35 to 45
(ii) greater than 45

If the expected number of students are greater than 200 whose score are less than 35, then school will be black listed. Is this school will be black listed, why?

SECTION – B

- 5(a). What is meant by a transcendental equation? Mention some methods to find the roots of nonlinear equations. 05

- 5(b). Find a root of $x \sin x - 1 = 0$ in the interval $[0, 2]$ by using the bisection method upto five steps. 10

- 5(c). Determine the real root of $x^3 - 6x^2 + 11x - 6.1 = 0$ by using the Newton-Raphson method starting with $x_0 = 3.5$ (three iterations). 10

- 5(d). What is meant by Numerical methods? Why should you study numerical methods? 10

- 6(a). Use Golden section search method to find the minimum of the following function: 10
 $f(x) = x^2 - 6x + 15$ between $x = 0$ and $x = 4$ upto three steps.

- 6(b). Fit a straight line to the following set of data by using least square method, 15

x	1	2	3	4	5
y	3	4	5	6	8

Hence estimate y for $x = 15$. Also plot the points and sketch the straight line.

- 6(c). The temperature of a metal strip was measured at various time intervals during heating and the values are given in the following table: 10

Time t (min)	1	2	3	4
Temperature T ($^{\circ}\text{C}$)	70	82	100	120

If the relationship between the temperature T and time t is of the form $T = be^{t/4} + a$, estimate the temperature at $t = 10$ min.

- 7(a). Solve the following system of linear equation by using LU decomposition method. 12

$$2x + y + z = 5, 3x + 5y + 2z = 15, 2x + y + 4z = 8$$

- 7(b). Apply Gauss-Seidel iterative technique to solve the following linear system of equations: 13

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

- 7(c). Determine the derivative of the function $f(x)$ tabulated below at the point $x = 1.1$: 10

x	1	1.2	1.4	1.6	1.8	2.0
$f(x)$	0	0.128	0.5440	1.2960	2.4320	4.0000

- 8(a). Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rules. 10

- 8(b). The following table gives the population of a town during the last six censuses. Estimate using any suitable interpolation formula, the increase in the population during the period from 1946 to 1948: 13

Year	1911	1921	1931	1941	1951	1961
Population (in thousand)	12	15	20	27	39	52

- 8(c). Apply Runge-Kutta (fourth order) method to approximate y , when $x = 0.1$ and $x = 0.2$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$. 12

--- End ---